

Prevalence and risk factors for diabetes mellitus among adults in Ghana: a systematic review and meta-analysis

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Received 20 May 2018; revised 26 July 2018; editorial decision 14 August 2018; accepted 19 September 2018

Background: The prevalence and risk factors for diabetes in Ghana vary from location to location, yet no study has been conducted to quantitatively synthesize the available studies.

Methods: The authors conducted a comprehensive literature search in Medline (PubMed), Embase, Cinahl, Web of Science, Scopus, African journals and grey literature. Two reviewers screened the articles for relevance, identified and selected studies based on inclusion and exclusion criteria. Methodological quality was independently assessed, using two validated assessment-of-bias tools. We explored study heterogeneity and performed a publication bias check.

Results: This search yielded a total of 624 articles, of which 17 studies were finally selected for the qualitative review and 12 studies included in the meta-analysis. The overall prevalence of diabetes mellitus among adult Ghanaians was high at 6.46% (95% CI: 4.66–8.26%) based on the inverse-variance random-effects model. Significant risk factors associated with diabetes were a family history of diabetes (OR: 3.46, 95% CI: 2.50–4.78), physical inactivity (OR: 3.06, 95% CI: 1.66–5.64) and age ≥ 40 years (OR: 2.36, 95% CI: 1.77–3.16).

Conclusion: Studies with high methodological quality provided sufficient evidence about diabetes prevalence among adults and the associated significant risk factors in Ghana.

Keywords: determinants, diabetes mellitus, Ghana, prevalence, risk factors, systematic review.

Introduction

Globally, the prevalence of diabetes has escalated comprehensively, with recent estimates of over 422 million adults aged 18 years and older, and an estimated attributable death rate of 1.5 million per year. Diabetes is projected to be the seventh leading cause of death by the year 2030,¹ with the numbers expected to increase to over 600 million by the year 2030.² More than two-thirds of the rise in the prevalence of diabetes is expected in low- and middle-income countries, including those in sub-Saharan Africa.³ The burden of the disease in Africa has risen significantly from an estimated 7.1 million people in the early 2000s to an estimated 18.6 million by 2030.⁴ Trends in the prevalence of diabetes in Ghana do not differ significantly from those observed in other sub-Saharan African countries, with earlier studies showing

a meagre prevalence rate of less than 0.02% of the adult population and recent figures ranging from 6.2% to 13.9%.^{5–8} These figures may not be alarming, as a sizeable proportion of individuals are left undiagnosed in the adult population.⁹

Risk factors driving the prevalence of diabetes differ between various continents, as well as within continents.¹⁰ Understanding and investigating the prevalence of and the peculiar risk factors for diabetes in sub-Saharan Africa will enable specific interventions to be targeted, in order to reduce the magnitude and burden of the disease. Although diabetes remains a public health issue posing a severe threat to Ghana's public health system, policymakers, researchers, and service providers are still unsure of the actual disease prevalence. This is mainly because the available evidence is based primarily on cross-sectional data, which tend to increase the

risk of underdiagnosis and bias. Additionally, most of the studies were conducted in different geographical locations of the country with highly variable prevalence rates and risk factors. Despite these limitations, no research has been undertaken to review the available studies systematically. To address this knowledge gap, the authors quantitatively synthesized results of independent studies to produce a more precise estimate of the prevalence of diabetes and its associated risk factors. The authors have also presented the highest level of evidence on the actual burden of diabetes in Ghana.

The objective of this review is to systematically identify, select and summarize the literature on studies conducted on prevalence and risk factors of diabetes in Ghana.

Methods

The most recommended guidelines, namely the Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) and Meta-analyses Of Observational Studies in Epidemiology (MOOSE), were followed as a guide for conducting the systematic review and meta-analysis of the selected studies.^{11,12}

Literature search

An extensive literature search in Medline (PubMed), Embase, Cinahl and Web of Science, with hand searching in Scopus, African journals and grey literature, was conducted. The search included articles published from 1 January 2000 to 20 March 2018. The search in the above-listed databases commenced by using keywords, MeSH terms, and a combination of MeSH terms and keywords based on the research question of interest. The MeSH terms used included 'Prevalence' [Mesh], 'Diabetes mellitus' [Mesh], 'Diabetes mellitus, Type 2' [Mesh], 'Ghana' [Mesh] and 'Risk factors' [Mesh]. The keywords used for the search were 'prevalence', 'diabetes prevalence', 'type 2 diabetes', 'T2D' or 'type II diabetes' or 'diabetes mellitus', 'risk factors', 'risk factor*', 'determinant*' and 'Ghana'. This comprehensive search focused on studies conducted on adult humans (18 years and older) with no restrictions on the language of publication of the studies. The reference lists of the retrieved articles were screened to identify additional relevant studies. Authors and experts who have, over the years, contributed significantly to the literature on diabetes were contacted to enquire about any additional published and unpublished studies. The search started on 6 February 2018, and the last search was completed on 20 March 2018. The final search strings/strategies for articles in the various databases are shown in Appendix S1 of the supplementary material. During the initial literature search, all forms of observational studies (case series, cross-sectional studies, case-control studies and cohort studies) were included.

Studies were included if diabetes, type 2 diabetes, gestational diabetes mellitus (GDM) or any form of diabetes was mentioned as one of the outcomes, including reviews on the question of interest. Additionally, studies that investigated the risk factors or determinants of diabetes mellitus, type 2 diabetes and GDM were included. The exclusion criteria were:

- case series and case reports;
- studies that did not indicate any type of diabetes as the outcome;
- studies conducted on non-Ghanaians below the age of 18 years.

For the quantitative synthesis (meta-analysis), studies that investigated the prevalence and/or risk factors of diabetes mellitus or type 2 diabetes among adults in Ghana were included. Also, cohort, case-control and cross-sectional studies were included. Additionally, studies that published measures of association, such as OR and 95% CI, as effect estimates and absolute effect estimates, specifically proportions, were included. Studies that examined the prevalence and risk factors of GDM in the quantitative synthesis were excluded.

Study selection and quality assessment

Two reviewers (MAB, ABT) independently provided quality assessment of the articles at all stages. At the initial step, the two reviewers independently scanned through the titles and abstracts of the articles retrieved from the databases. The relevance of the studies was assessed, and the articles selected were further screened. Seventeen articles—six case-control studies and 11 cross-sectional studies—were finally extracted from the screening stage. The US Preventive Services Task Force (USPSTF) Quality Rating Criteria assessment of bias for case-control studies and the risk-of-bias tool for prevalence studies by Hoy et al. (2012) were used to perform the quality assessment of the individual studies.^{11,13} Using the USPSTF assessment-of-bias tool, each reviewer independently assessed and graded each of the six case-control studies as 'good', 'fair' or 'poor'.¹⁴ The assessment-of-bias tool for prevalence studies was used by the two authors to evaluate the 11 cross-sectional studies independently and to grade them as 'low risk', 'high risk' or 'unclear risk of bias'.

An independent reviewer was involved in the review to black out names of the authors, the title of the study and institution of affiliation before evaluating the individual studies. This approach provided blinding and resolved any disagreement that arose from the assessment of the articles by the two reviewers.

Data extraction

Using some generic items and relevant information, a standardised form was generated and used for data extraction. The two authors independently performed the entire data extraction process simultaneously and addressed some recognised differences regarding selected risk factors, which were reconciled by agreement. The form was pre-tested on four selected studies before adopting it for the entire extraction process. The following information was extracted by using the standardised form: author's name and year of publication, study design, sampling technique, study location, cases, risk factors, outcomes and results (see Table 1 and Table S1 in the supplementary material). Also, effect estimates of the relative measure (OR) and their corresponding 95% CIs were extracted from appropriate studies.

Statistical analysis

The principal summary measures for the 12 studies included in the quantitative synthesis were OR and proportions. In all, two different meta-analyses were conducted. First was a meta-analysis of the prevalence of diabetes and the second summed up evidence on the risk factors for diabetes using their respective effect estimates and 95% CIs. The Review Manager (RevMan)

Table 1. Summary of findings of studies included in the review for investigating the prevalence of diabetes in Ghana

Article	Population/participants	Study type/design	Sampling technique	Study location	Cases	Outcomes	Results
Amoah et al., 2002 ⁷	4733 Ghanaians aged 25 years and above from some selected areas in the Greater Accra region participated in the survey	Cross-sectional study	A stratified two-stage random cluster sampling	Greater Accra Region (selected communities were Labadi/Cantoments Teshie and Danfa/Abokobi)	300 subjects were identified as having diabetes	Prevalence Type 2 diabetes mellitus (DM2)	The crude prevalence of diabetes was 6.3% (5.61–6.99%) and the age-adjusted prevalence was 6.4%. Participants aged 64 years and older recorded the highest prevalence of 13.6. The mean age was 44.3±14.7
Quaicoe et al., 2017 ¹⁵	Adults aged 18–64 comprising 75 cases and 151 controls participated in the study	Unmatched case-control study	Multistage sampling technique	Volta Region (Hohoe municipality)	75 cases (patients with diabetes attending diabetes clinic)	Prevalence and risk factors of DM2	The study recorded diabetes prevalence of 70.7% in cases, 8.6 % (4.7–12.5%) in controls. Mean age of participants was 58.0±10.8. Mean age of cases was 59.89±9.92
Vuvor et al., 2011 ²⁷	288 men and 309 women aged between 36 and 95 years participated in the study	Cross-sectional study	A two-staged sampling strategy	Greater Accra (selected urban and peri-urban communities)	23 participants with diabetes, comprising 10 males and 13 females determined by fasting blood glucose	Prevalence and risk factors of DM	Prevalence of DM among subjects was 3.9% (2.3–5.5%)
Cook-Huynh et al., 2012 ¹⁶	Adults aged 18 years and older. 326 adults were sampled (94 males, 232 females)	Cross-sectional study, observational study	A two-tier random sampling approach	Rural Adankwame community, Ashanti Region, Ghana	DM and hypertension. DM diagnosis was based on the WHO criteria of a fasting blood sugar (FBS) level 126 mg/dL or above OR a random blood sugar level of greater than or equal to 200 mg/dL	Prevalence of hypertension and diabetes	The overall prevalence of DM in this sample population was 8.0% (5.0–11.0%)
Darkwa, 2011 ²⁹	Health facilities in the municipality for caring for diabetic patients	Facility-based study	Random sampling approach	Cape Coast Metropolis, Central region	DM	Prevalence of DM and resource availability	Inadequate resources for managing diabetes in the municipality
Tarekegne et al., 2018 ³⁰	4167 Adults aged 50 years and older sampled from the SAGE data	Cross-sectional study based on WHO SAGE wave 1	Stratified random sampling employed by WHO	Conducted in various parts of Ghana by WHO	122 participants self-reported diagnosis of DM or DM status	Various characteristics associated with self-reported DM	3% (2.4–3.6%) Prevalence rate of self-reported DM diagnosis. 3% for females, 4% for males
Sarfo-Kantanka et al. (2014) ⁸	1292 healthy individuals aged greater than or equal to 40 years were screened for diabetes and hypertension	Cross-sectional study	Sampling was based on individuals' willingness to participate	Ashanti Region (Kumasi municipality, where five sub-metropolises were selected)	Participants diagnosed with diabetes after the screening. Diagnosis based on WHO guidelines of FBS greater than or equal to 7.0 mm/L	Diabetes prevalence	Prevalence of diabetes in healthy individuals was 13.9% (12.0–15.8%)
Frank et al., (2013) ²⁸	1221 adults comprising (542 cases, 679 controls) residing in urban Ghana participated in the study	An unmatched case control study		Ashanti Region (Kumasi)	542 individuals with diagnosis of DM2. Physician diagnosis based on WHO guidelines	Obesity and risk of DM2	Measures of central (including waist-hip circumference, waist hip-ratio), rather than obesity increases one's likelihood of developing DM2
Gato et al., 2017 ¹⁷	482 adults aged 18 years and older were sampled from Cape Coast Metropolis to participate in the study	Cross-sectional study	A convenience sampling technique	Central region (Cape Coast Metropolis)	Individuals with self-reported diagnosis of diabetes	Self-reported diagnosis and prevalence of DM	8.3% (5.8–10.8%) of the sampled individuals reported ever diagnosis of diabetes The association of diabetes and socioeconomic position (SEP)

Continued

Table 1. Continued

Article	Population/participants	Study type/design	Sampling technique	Study location	Cases	Outcomes	Results
Addo et al., 2017 ¹⁸	5290 Ghanaians aged 25–70 from the Obesity and Diabetes among African Migrants (RODAM) study participated in the study	Multicentre cross-sectional study	Random sampling approach was adopted to select respondents from the list of 30 census enumeration areas in the Ashanti region	Ghanaians residing in rural and urban Ghana in the Ashanti Region, the Netherlands (Amsterdam), UK (London) and Germany (Berlin)	Participants with ever diagnosis of diabetes and obesity. The diagnostic criteria for diabetes were defined as fasting plasma glucose ≥ 7.0 mmol/L or self-reported diabetes	Prevalence of diabetes and obesity	(educational level and occupation class) varied in rural Ghana compared with urban settings in Ghana and Europe
Asare-Anane et al. (2014) ¹⁹	A total of 200 pregnant women attending ante-natal care (100 pregnant women with gestational diabetes mellitus [GDM] and 100 pregnant women without GDM) participated in the study	Case-control study		Pregnant women attending ante-natal care at the Maternity Unit of the Korle-Bu Teaching Hospital and the National Diabetes Management and Research Centre, in Accra, Ghana	100 pregnant women with diagnosis of GDM. Diagnostic criteria: glycated haemoglobin of less than 6.5%	Risk factors of GDM	The mean age for the cases was 33, and that of the controls was 35.06 years
Oppong et al., 2015 ²⁰	Pregnant women within their first 20–24 weeks attending their first prenatal clinic were enrolled into the study	Cross-sectional study		Prenatal clinic at Korle-Bu Teaching Hospital, Accra, Ghana	37 (9.3%) out of 399 women were having or physician diagnosed GDM based on WHO guidelines	Determining the burden of gestational diabetes among pregnant women in Accra	9.3% (6.4–12.2%) of pregnant women screened had GDM
Danquah et al., 2012 ²¹	1466 individuals recruited from the diabetes and hypertension clinic between August 2007 and June 2008	Case-control study with the controls selected from the community through voluntary participation		Komfo Anokye Teaching Hospital (KATH), diabetes and hypertension clinic, Kumasi.	DM2 was defined as fasting plasma glucose ≥ 7 mmol/L	Examining factors associated with DM2 and hypertension	Majority of the DM2 patients were females (n=675) with mean age of 55 years. DM2 mellitus affected obese patients with low socioeconomic status
Agbogli et al., 2017 ²²	113 Adults 18 years and older were sampled from Oforikrom Sub-Metro in Kumasi	Cross-sectional study	Random sampling approach was used to sample individuals from two communities	Oforikrom Sub-Metro, Kumasi Metropolis	DM2 patients using the WHO FBS guidelines	Prevalence and associated risk factors of DM2	The study found 3.5% (0.1–6.9%) observed prevalence of DM
Fiagbe et al., 2017 ²³	Adults aged 18 years and older residing in the Hohoe municipality	Hospital-based case- and sex-matched case-control study		Hohoe municipality, Volta Region	70 DM2 patients. Diabetes is diagnosed with a blood glucose value of ≥ 200 mg/dL or Random blood glucose levels (RBS) > 11.0 mmol/L	Prevalence, control and risk factors associated with DM2 among adults in the Hohoe municipality	Prevalence of uncontrolled DM2 was 78.6%
Gatimu et al., (2016) ²⁴	4135 Adults aged 50 years and older sampled from Study of Ageing and Adult Health (SAGE) Wave 1 formed the study population	Cross-sectional study based on SAGE data	Multistage stratified cluster design	A survey conducted by WHO across the various parts of Ghana (nationwide)	Self-reported doctor diagnosis of diabetes	Prevalence and determinants of self-reported diagnosis of diabetes	The weighted prevalence of diabetes among the older adults was 3.95% (3.35–4.55%). Diabetes prevalence for females: 2.16% (1.69–2.76%), males: 1.73% (1.28–2.33%)
Gudjinu & Sarfo, 2017 ²⁵	A total of 136 participants (48 cases, 88 controls) were recruited. Participants were between 35 and 62 years old	Unmatched case-control study		Ho municipality, Ghana	Those diagnosed with DM2. The diagnostic criteria were based on the laboratory testing for fasting blood glucose and fasting blood lipid	Assessing the risk factors of DM2	

version 5.3 software (Copenhagen: The Nordic Cochrane Centre) was used for data analyses, which generated the forest and funnel plots. A random-effect model with generic inverse-variance method was used due to suspected heterogeneity across selected studies, such as the reported prevalence rates, diabetes type and risk factors for diabetes. Funnel plots were generated to visually inspect the presence or absence of publication bias in this review. Sensitivity and sub group analyses were performed to assess the robustness of the study results in the presence of possible variations in the study results.

Setting

Ghana is a country on the west coast of West Africa that shares boundaries with Togo to the east, La Cote d’Ivoire to the west, Burkina Faso to the north and Gulf of Guinea to the south. The country covers an area of 238 500 km², with an estimated population of 25 199 609 as of July 2013. Ghana is divided into 10 administrative regions, namely Ashanti, Eastern, Western, Volta, Central, Brong Ahafo, Northern, Upper East, Upper West and Greater Accra, where the capital city Accra is located.²⁶ (See Figure S1 in the supplementary material.)

Results

Literature search results

The literature search conducted in the four main databases and by hand-searching in some journals and grey literature (theses) yielded 624 articles (PubMed=50, Embase=98, Cinahl=381, Web of Science=58, hand-searching in journals=35, grey literature [theses]=2). No additional study was found after scanning through the reference lists of the retrieved studies. Duplicates were removed from the retrieved articles using the reference manager software, leaving 575 articles. Here, the titles and abstracts of the 575 articles were screened for relevance by the authors MAB and ABT. Out of the 575, 64 articles were retrieved and downloaded for full-text review. All 64 articles were assessed and 47 excluded for not meeting the inclusion criteria, leaving 17 articles for the qualitative assessment (see Figure 1).

Summary of included studies

The final 17 articles comprised six case-control and 11 cross-sectional studies. The retrieved studies were conducted in some selected communities of Accra (in the Greater Accra Region),^{7,19,20,27}

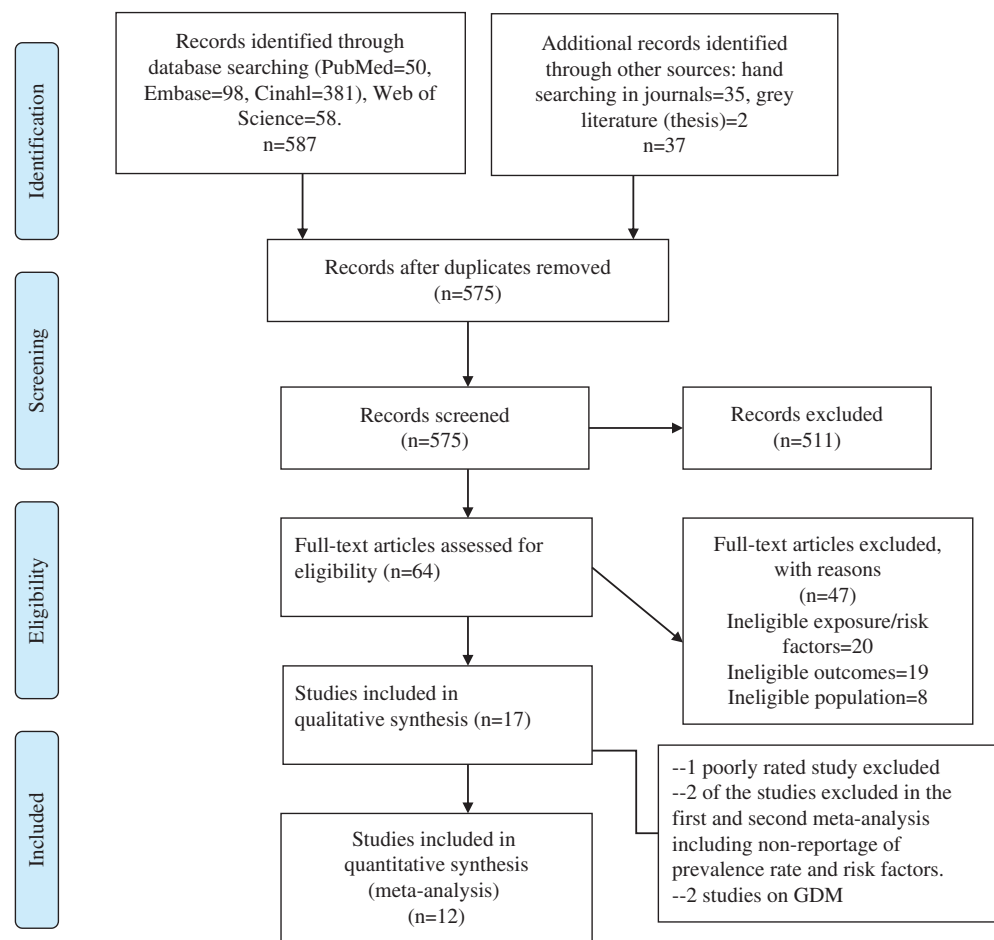


Figure 1 Preferred reporting item for systematic review and meta-analysis flow diagram. GDM: gestational diabetes mellitus. The left rectangles shows the four main stages we employed in searching for the studies (From Identification stage to inclusion of articles). The connected boxes in the flow chart shows the search results through to the final included studies.

Kumasi (in the Ashanti Region),^{18,21,22,28} Adankwame (in the Ashanti Region),¹⁶ Ho (in the Volta Region),^{23,25} Hohoe (in the Volta Region),¹⁵ Cape Coast (in the Central Region)^{17,29} and two nationwide surveys.^{24,30} The year of publication of the selected studies ranged from 2002 to 2017. Most of the studies were based on cross-sectional data, and participants were mostly sampled from preferred communities within their respective regions. Studies that investigated the risk factors of diabetes reported the main measure of association as OR. Most of these studies summarised patients' characteristics in a tabular form, adjusted for relevant covariates in their multivariate analysis using the binary logistic regression model. All 17 studies were reviewed qualitatively, and 12 out of the 17 studies were reviewed quantitatively using meta-analysis (see Tables 1 and S1 in the supplementary material).

The evaluation of the quality of each study based on the overall quality ratings is summarized in Tables S2 and S3 in the supplementary material. Generally, the papers selected for the review were of 'good' quality, with a low risk of bias in study methodology. Of the 17 studies included, only one was graded as at 'high risk' of bias due to methodological flaws, including non-representativeness of the target population, non-usage of random sampling approach in selecting study participants, high non-response rate, no accepted diagnostic method utilized and mode of data being dissimilar across subjects.²⁹ The case-control studies included in this review were rated as 'good' with a low risk of bias in assessing the methodological quality of the studies. Only one out of the six case-control studies was rated as 'fair' quality. Overall, the risk of bias was minimal across the studies reviewed.

The majority of the studies included^{8,16,18,21-23,25,27,28} defined diabetes diagnosis according to the WHO guidelines/criteria of fasting plasma glucose ≥ 7 mmol/L or 126 mg/dL, and/or a random blood sugar level of greater or equal to 11.1 mmol/L/200 mg/dL.³¹ Studies included were conducted in four out of the 10 administrative regions of Ghana, namely Central Region, Ashanti Region, Greater Accra Region and Volta Region. Most of the studies were conducted in the Ashanti and Greater Accra regions of Ghana.^{7,8,16,18-20,22,25,27,28} Two nationwide studies using the WHO SAGE wave 1 data were included.^{24,30} These primary studies

including the two national surveys on diabetes could be representative of the entire Ghanaian adult population, and hence, the synthesis of these individual studies into one prevalence and risk factor estimate could be representative as well. In all the reviewed studies, diabetes prevalence ranged from 3.0% to 13.9%, with the highest recorded in Kumasi (13.9%)⁸ followed by the next highest (9.3%) in Accra⁷ in the Greater Accra Region and the lowest (3%) reported by one of the nationwide survey studies using the WHO SAGE data.³⁰ The common risk factors for diabetes mellitus included a family history of diabetes, obesity/overweight, older age (40 years and older) and physical inactivity. Other risk factors identified in some studies include high waist-to-hip ratio, low consumption of fruit and vegetables, low high-density lipoprotein cholesterol, and a secondary and tertiary level of education.^{8,25,28,30} The study by Fiagbe et al.²³ identified vigorous physical activity work as a risk factor with an OR of 3.67 (95% CI: 1.58–8.50). (Refer to Tables S2 and S3 in the supplementary material.)

Meta-analysis

Prevalence of diabetes mellitus

The authors conducted meta-analyses on 12 studies in total, comprising four case-control studies and eight cross-sectional studies, after their quality was assessed with the two quality assessment-of-bias tools. Two separate meta-analyses were conducted: one on studies that reported the prevalence of diabetes and the second on studies that reported comparable risk factors for diabetes mellitus. The main measure of prevalence was in proportion (percentage), and their respective 95% CIs were computed with the RevMan calculator. The authors adopted the generic inverse-variance method to estimate the weights of each of the studies based on each study's summary measure. Nine studies were included in determining the pooled effect of prevalence of diabetes among adults in Ghana.^{8,15-17,22,24,27,30} From Figure 2, the combined effect estimate of diabetes prevalence with 95% CI is 6.46% (4.66–8.26%) with $p < 0.00001$. This gives an overall significance of diabetes prevalence of approximately 6.5% among Ghanaian adults, which is

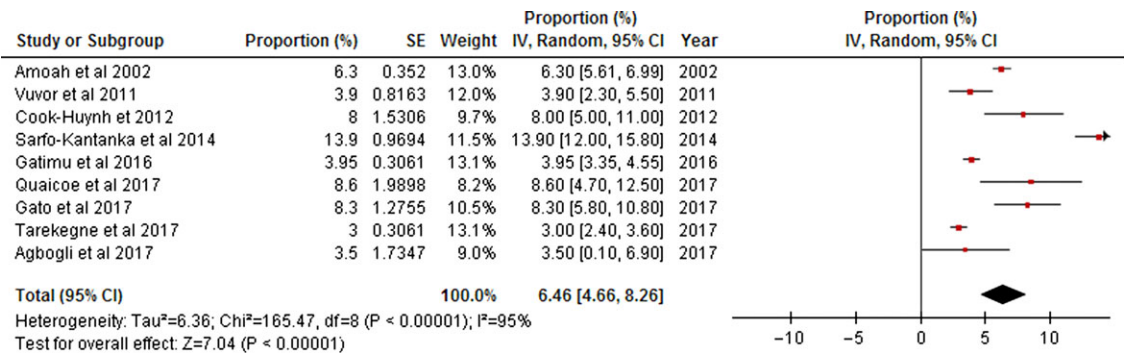


Figure 2. Forest plot of the prevalence of diabetes among adults in Ghana. The first column shows the identities of the included studies. Studies are represented by Surnames and year of publication. The second and third columns shows the prevalence proportion (%) and the standard errors (SE) of each of the prevalence proportions respectively. The fourth column of the figure gives the weight (in %) which indicates the influence an individual study has had on the pooled result. Column 6 displays the year of publication for each study. The seventh column visually displays the study results. The tiny boxes in the middle of the horizontal lines show the effect estimates from the single studies, while the diamond shows the pooled result. The horizontal lines through the boxes illustrate the length of the confidence interval. The vertical line is the line of no effect. The X-axis shows the scale of effect estimates. The fifth column displays exactly the same information (in numerical format) as contained in the diagram in column 7.

expected to fall within the closed interval of 4.66% to 8.26% with substantial heterogeneity of $I^2=95\%$.

Subgroup and sensitivity analysis

To identify the source of the significant heterogeneity among the studies, subgroup analysis was performed by subgrouping studies, based on the findings reported from the country's administrative regions or locations. The first subgroup pooled studies with reported diabetes prevalence from the Ashanti Region (located in the southern belt of Ghana). The second pooled studies that reported findings from areas/regions along the coastal belt of the country, specifically Volta, Greater Accra and Central regions of Ghana. Three studies^{8,16,22} were classified under the 'Ashanti Region' subgroup and three from the second subgroup.

Studies classified under the first subgroup recorded a pooled prevalence rate of 8.59 (95% CI: 2.38–14.80, $p<0.00001$) with significant heterogeneity ($I^2=94\%$) (see Figure 3). The overall diabetes prevalence among studies in the second subgroup was 7.10% (95% CI: 5.54–8.65, $p<0.0001$). Studies conducted along the coastal belt of Ghana recorded a similar prevalence rate of 7.1% (with I^2 of 41%).

Risk factors for diabetes mellitus

The second meta-analysis was performed on selected studies that reported comparable risk factors across studies. The four comparable risk factors identified and considered as individual subgroup headings were a family history of diabetes, physical inactivity, obesity/overweight and older age (40 years and older).^{15–17,21,24} Results of the study synthesis are summarised in the forest plot of Figure 4. Three studies reported family histories of diabetes as a significant risk factor for diabetes, with pooled effect estimate of 3.46 (95% CI: 2.50–4.78), with no significant

heterogeneity among included studies.^{16,21,23} Compared with individuals with no family history of diabetes, those with a family history of diabetes were 3.46 times more likely to be diagnosed with diabetes, and this association was significant. All three studies reported similar increasing likelihood of developing diabetes with a family history of diabetes (with I^2 of 0%). Also, older adults (40 years and older) recorded an increased likelihood of developing diabetes (OR: 2.36, 95% CI: 1.77–3.16, $p<0.00001$) compared with younger adults below the age of 40 years. Studies included similar effect estimates of age reported as a risk factor with $I^2=0\%$ (see Figure 4). Additionally, from Figure 4, physical inactivity was one of the significant risk factors reported by the three studies, with overall effect estimate of OR: 3.06 (95% CI: 1.66–5.64), and p -value of 0.0004. The effect estimate was similar across the three studies ($I^2=34\%$). Finally, obese and overweight individuals were found to have an increased likelihood (OR: 2.12, 95% CI: 1.21, 3.72) of developing diabetes mellitus compared with those with normal body mass index.^{8,17,23–25}

Publication bias check

The funnel plot in Figure S2 in the supplementary material was employed to assess and detect publication bias visually. By inspecting the funnel plot, there seems to be some significant level of symmetry among the eight studies after excluding one.^{15–17,22,24,27,30} This means that there could be no significant publication bias among studies included in the synthesis of diabetes prevalence. (See Figure S2 in the supplementary material.)

Discussion and conclusion

The primary aim of this study was to summarize the available literature on the prevalence and risk factors of diabetes mellitus

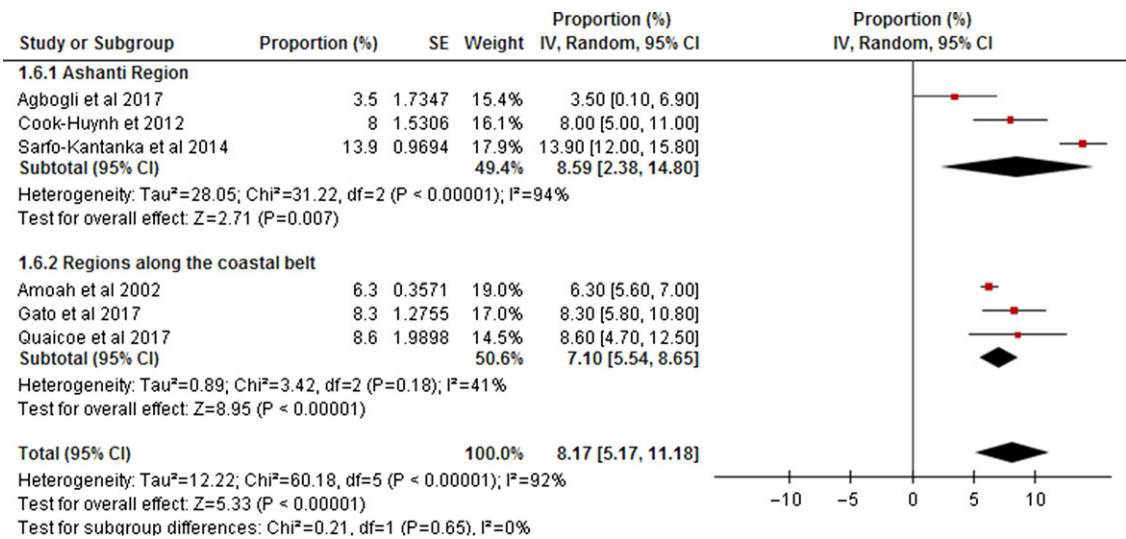


Figure 3. Forest plot of the prevalence of diabetes among Ghanaian adults (subgroup analysis). The first column shows the identities of the included studies. Studies are represented by Surnames and year of publication. The second and third columns shows the prevalence proportion (%) and the standard errors (SE) of each of the prevalence proportions respectively. The fourth column of the figure gives the the weight (in %) which indicates the influence an individual study has had on the pooled result. Column 6 visually displays the study results. The tiny boxes in the middle of the horizontal lines show the effect estimates from the single studies, while the diamond shows the pooled result. The horizontal lines through the boxes illustrate the length of the confidence interval. The vertical line is the line of no effect. The X-axis shows the scale of effect estimates. The fifth column displays exactly the same information (in numerical format) as contained in the diagram in column 6.

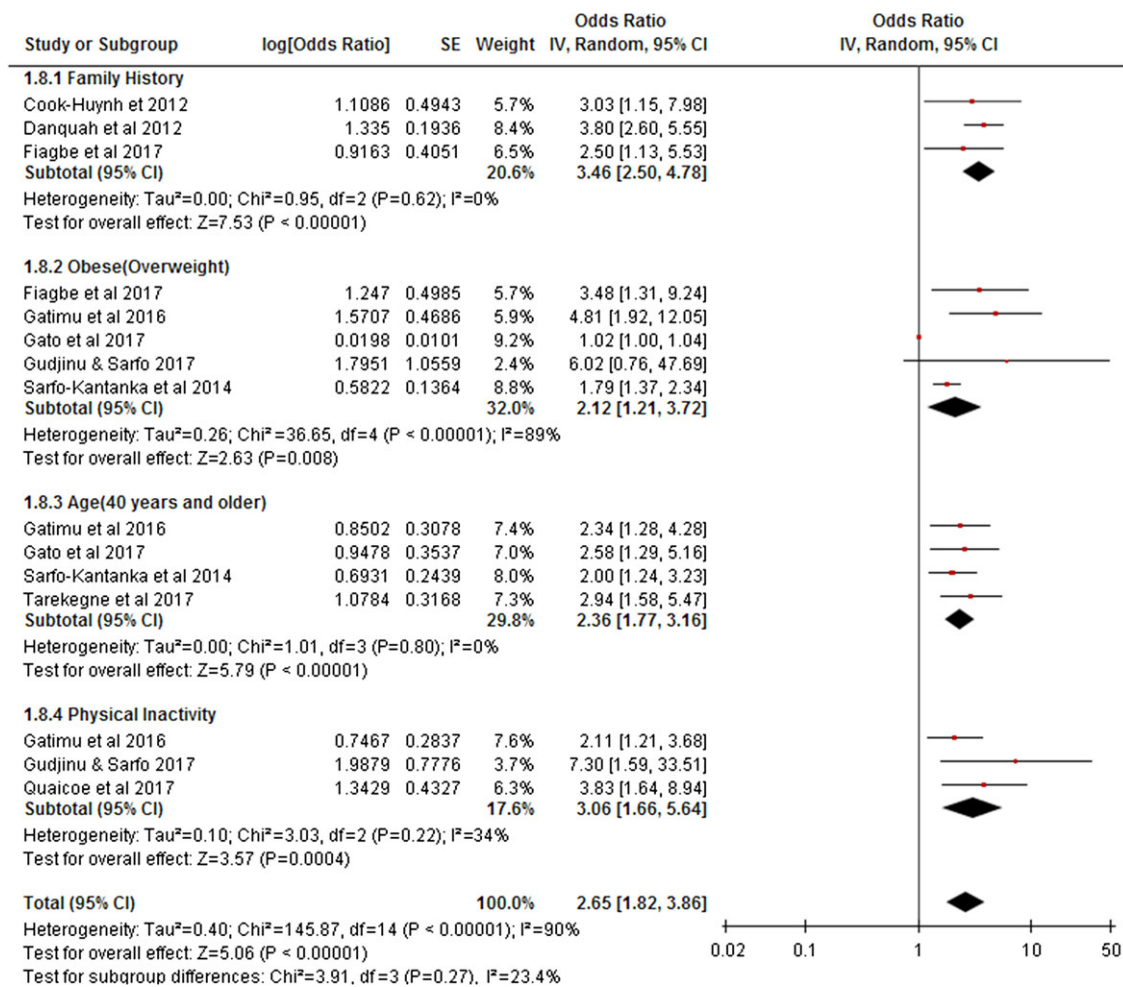


Figure 4. Forest plot for risk factors and determinants of diabetes prevalence among Ghanaian adults. The first column shows the identities of the included studies. Studies are represented by Surnames and year of publication. The second and third columns shows the log of the estimates (log odds) and their corresponding standard errors (SE) for each included study. The fourth column of the figure gives the weight (in %) which indicates the influence an individual study has had on the pooled result. The Sixth column visually displays the study results. The tiny boxes in the middle of the horizontal lines show the effect estimates from the single studies, while the diamond shows the pooled result. The horizontal lines through the boxes illustrate the length of the confidence interval. The vertical line is the line of no effect. The X-axis shows the scale of effect estimates. The fifth column displays exactly the same information (in numerical format) as contained in the diagram in column 6.

among Ghanaian adults (aged 18 years and older) through systematic means of identifying, selecting and critically appraising the literature. To the best of the authors' knowledge, this study is the first comprehensive systematic review and meta-analysis of this research question. Currently, studies conducted on diabetes prevalence in the country vary from location to location, with the majority based on cross-sectional data, contributing to underdiagnosis of the condition. To date, the actual prevalence of diabetes in Ghana is unclear, and the prevalence of the disease among adults unknown. Synthesizing studies conducted on diabetes prevalence draws us close to what might be the real state of the disease prevalence. To achieve this, strict inclusion and exclusion criteria were adopted in the identification and selection of studies to be included in this review. This study included studies that investigated prevalence and risk factors of diabetes among Ghanaian adults, including type 2 diabetes and GDM. Studies were excluded if they were not conducted among

adult Ghanaians. The quality-assessment tools rated studies generally as having a low risk of bias. The majority of these studies followed high standards of conducting research and, hence, were sufficient to draw broad conclusions on the overall prevalence and significant risk factors for diabetes mellitus.

Our findings provide reliable evidence of the near actual prevalence of diabetes in Ghana and the significant risk factors that need to be targeted in order to reduce the burden of this condition. Although the current evidence indicates that 540 thousand Ghanaians are living with diabetes, a substantial proportion of Ghanaians are unaware that they live with the disease and remain undiagnosed, as with most other countries worldwide.^{1,32} However, synthesizing the various epidemiological studies conducted in Ghana about the condition using hospital-based data and cross-sectional data will give an accurate picture of the disease prevalence and its associated significant risk factors. Although studies reviewed were conducted

across the country, the majority of them were executed at specific locations in the country. Aside from the two nationwide cross-sectional surveys, the majority of the studies were conducted in four administrative regions of the country, which seems fairly representative.

The prevalence rates differed from study to study, depending on the location where the study was conducted. This result presupposes that there exists geographical variation of diabetes prevalence among adults in Ghana, which needs to be critically examined for policymakers to channel more resources into areas where the disease is endemic. This finding is corroborated by conclusions from China, which showed significant geographical variation in the prevalence of the disease.³³ The reason for the difference in diabetes prevalence within a country can be attributed to differential exposures to risk factors for diabetes, such as urbanisation, poor dietary practices and obesity, as well as the genetic make-up of individuals.^{34,35} The overall prevalence of diabetes of 6.46%, which is expected to reflect the current burden of diabetes among adults in the country, is similar to the worldwide prevalence of the disease as currently stated.¹

The studies included identified significant risk factors for diabetes mellitus, based on their pooled effect estimates, as family history of diabetes, physical inactivity and older age (40 years and older) with significant homogeneity. These findings are somewhat consistent and in agreement with studies that identified family history, low physical activity, increasing age and obesity as significant determinants of diabetes mellitus.^{4,36–39}

The following policy measures from this study can be deduced. First and foremost, the overall prevalence of diabetes seems to be high, especially in the southern part of the country, and specifically in Kumasi in the Ashanti region, Cape coast in Central and Accra in the Greater Accra regions (see Figure S1). This indicates that there could be geographical variations in diabetes prevalence. The current prevention and management programs targeted at the condition are based on an assumption of spatial homogeneity, which might contradict the unknown diabetes prevalence variabilities. Therefore, investigating how diabetes prevalence varies geographically is essential for efficient allocation of resources for management of the conditions and, consequently, provides clues to understanding the primary determinants of the disease. Therefore, spatial analysis of diabetes prevalence in Ghana as an area for future research is recommended.

Secondly, there is the need to educate adults on living a healthy lifestyle to limit their risk of developing diabetes and other non-communicable diseases. Additionally, education should accentuate the relevance of taking a healthy diet (including fruit and vegetables), engaging in regular physical activities and seeking proper medical attention, especially among older adults aged 40 years and above.

One significant strength of this review was the fact that studies included were of good methodological quality with low risk of bias, which provided the best of evidence of diabetes prevalence among adults in Ghana and its associated significant risk factors. Limitations of this study include:

- highly variable prevalence rates of diabetes;
- slightly different populations and their comparators and somewhat different risk factors, which made pooling a bit difficult.

Notwithstanding, this study is the highest level of evidence on studies conducted on diabetes prevalence and its risk factors from 2002 to 2017. This will serve as a robust and reliable document for policymakers and healthcare providers to be informed about the state of the burden of the disease and associated risk factors among Ghanaian adults.

Supplementary data

Supplementary data are available at *International Health online* (<http://inthehealth.oxfordjournals.org>).

Authors' contributions: MAB, OSK, ABT, BE and JCM conceived the study and designed the study protocol. MAB and OSK developed the search strategy and conducted the search for studies. MAB, ABT and JCM conducted the screening and study selection of cases. Data extraction was performed by MAB, OSK, ABT, BE and JCM. Data analysis and interpretation were carried out by MAB, OSK, ABT, BE and JCM. MAB, OSK, ABT, BE and JCM drafted the manuscript and critically revised the manuscript for intellectual content. MAB is the guarantor of the paper. All authors read and approved the final version of the manuscript.

Acknowledgements: None to declare.

Funding: None

Competing interests: Authors declare no competing interests.

Ethical approval: Not required

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